

Tree Model Code

```
# Sets seed for reproducibility  
set.seed(42)
```

```
# Reads in libraries  
library(rpart)  
library(rpart.plot)  
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.5.3
```

```
## randomForest 4.7-1.2
```

```
## Type rfNews() to see new features/changes/bug fixes.
```

```
# Reads in data frame  
df <- read.csv("insurance.csv")  
  
# Creates log transformation of charges  
df$log_charges <- log(df$charges)  
  
# Creates training and testing sets  
n <- nrow(df)  
train_index <- sample(1:n, size = 0.7 * n)  
  
df_train <- df[train_index, ]  
df_test  <- df[-train_index, ]  
  
# Prints header of data frame  
head(df)
```

```
##   age    sex    bmi children smoker   region   charges log_charges  
## 1  19 female 27.900         0    yes southwest 16884.924    9.734176  
## 2  18   male 33.770         1     no southeast  1725.552    7.453302  
## 3  28   male 33.000         3     no southeast  4449.462    8.400538  
## 4  33   male 22.705         0     no northwest 21984.471    9.998092  
## 5  32   male 28.880         0     no northwest  3866.855    8.260197  
## 6  31 female 25.740         0     no southeast  3756.622    8.231275
```

```
# Creates regression tree with all predictors  
rt1 <- rpart(log_charges ~ smoker + age + bmi + children + region + sex,  
             data=df_train, method="anova")
```

```
# Prints model summary
summary(rtl)
```

```
## Call:
## rpart(formula = log_charges ~ smoker + age + bmi + children +
##       region + sex, data = df_train, method = "anova")
## n= 936
##
##          CP nsplit rel error   xerror   xstd
## 1 0.43381990    0 1.0000000 1.0023041 0.03866529
## 2 0.25925166    1 0.5661801 0.5685257 0.02482450
## 3 0.04234692    2 0.3069284 0.3194009 0.02180424
## 4 0.03353305    3 0.2645815 0.2752341 0.02100894
## 5 0.02990710    4 0.2310485 0.2536395 0.01941204
## 6 0.01075742    5 0.2011414 0.2186641 0.01954893
## 7 0.01000000    6 0.1903839 0.2141097 0.02060618
##
## Variable importance
##   smoker      age children      bmi   region
##      52       39      4       4       1
##
## Node number 1: 936 observations,   complexity param=0.4338199
##   mean=9.115445, MSE=0.8221967
##   left son=2 (747 obs) right son=3 (189 obs)
##   Primary splits:
##     smoker   splits as LR,      improve=0.433819900, (0 missing)
##     age      < 35.5   to the left, improve=0.226033000, (0 missing)
##     children < 1.5    to the left, improve=0.027968210, (0 missing)
##     bmi      < 35.88  to the left, improve=0.016811030, (0 missing)
##     region   splits as RLLL,    improve=0.001453492, (0 missing)
##
## Node number 2: 747 observations,   complexity param=0.2592517
##   mean=8.815036, MSE=0.544556
##   left son=4 (268 obs) right son=5 (479 obs)
##   Primary splits:
##     age      < 32.5   to the left, improve=0.490467200, (0 missing)
##     children < 1.5    to the left, improve=0.055522360, (0 missing)
##     bmi      < 35.88  to the left, improve=0.016683860, (0 missing)
##     region   splits as RRLL,    improve=0.013072950, (0 missing)
##     sex      splits as RL,      improve=0.006867174, (0 missing)
##   Surrogate splits:
##     bmi < 17.83   to the left, agree=0.645, adj=0.011, (0 split)
##
## Node number 3: 189 observations,   complexity param=0.0299071
##   mean=10.30278, MSE=0.1530972
##   left son=6 (89 obs) right son=7 (100 obs)
##   Primary splits:
##     bmi      < 30.1   to the left, improve=0.795420500, (0 missing)
##     age      < 43.5   to the left, improve=0.110830400, (0 missing)
##     region   splits as LLRR,    improve=0.026738610, (0 missing)
##     children < 1.5    to the left, improve=0.007778093, (0 missing)
##     sex      splits as LR,      improve=0.004357908, (0 missing)
##   Surrogate splits:
```

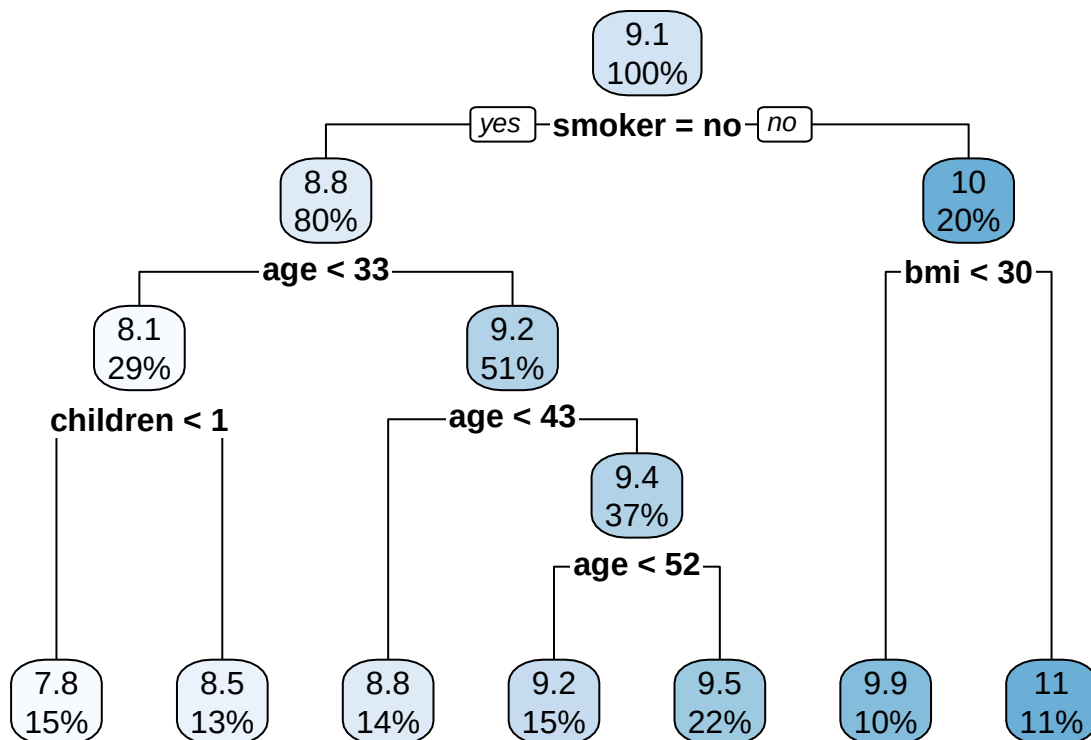
```

##      region  splits as  LLRR,          agree=0.598, adj=0.146, (0 split)
##      sex     splits as  LR,           agree=0.540, adj=0.022, (0 split)
##      age     < 21.5     to the left,  agree=0.534, adj=0.011, (0 split)
##      children < 2.5     to the right,  agree=0.534, adj=0.011, (0 split)
##
## Node number 4: 268 observations,      complexity param=0.03353305
## mean=8.124117, MSE=0.4624146
## left son=8 (143 obs) right son=9 (125 obs)
## Primary splits:
## children < 0.5      to the left,  improve=0.208237200, (0 missing)
## age < 22.5          to the left,  improve=0.149463600, (0 missing)
## region splits as  RRLl,          improve=0.049802280, (0 missing)
## bmi < 23.46         to the left,  improve=0.018358330, (0 missing)
## sex splits as  RL,           improve=0.008997053, (0 missing)
## Surrogate splits:
## age < 24.5          to the left,  agree=0.664, adj=0.280, (0 split)
## region splits as  LRLR,          agree=0.552, adj=0.040, (0 split)
## sex splits as  LR,           agree=0.552, adj=0.040, (0 split)
## bmi < 28.215        to the left,  agree=0.545, adj=0.024, (0 split)
##
## Node number 5: 479 observations,      complexity param=0.04234692
## mean=9.201604, MSE=0.1739923
## left son=10 (132 obs) right son=11 (347 obs)
## Primary splits:
## age < 42.5          to the left,  improve=0.391028000, (0 missing)
## bmi < 34.785         to the left,  improve=0.020083240, (0 missing)
## sex splits as  RL,           improve=0.012344180, (0 missing)
## region splits as  RRLl,          improve=0.007727859, (0 missing)
## children < 0.5      to the right,  improve=0.007035719, (0 missing)
## Surrogate splits:
## bmi < 18.0975        to the left,  agree=0.729, adj=0.015, (0 split)
##
## Node number 6: 89 observations
## mean=9.932874, MSE=0.04433209
##
## Node number 7: 100 observations
## mean=10.63199, MSE=0.01974027
##
## Node number 8: 143 observations
## mean=7.833995, MSE=0.3118515
##
## Node number 9: 125 observations
## mean=8.456018, MSE=0.4282089
##
## Node number 10: 132 observations
## mean=8.778695, MSE=0.09123539
##
## Node number 11: 347 observations,      complexity param=0.01075742
## mean=9.36248, MSE=0.1115564
## left son=22 (145 obs) right son=23 (202 obs)
## Primary splits:
## age < 51.5          to the left,  improve=0.213863100, (0 missing)
## sex splits as  RL,           improve=0.023005920, (0 missing)
## bmi < 23.6275        to the left,  improve=0.017386360, (0 missing)

```

```
##      region splits as RRL,      improve=0.011560880, (0 missing)
##      children < 2.5 to the left, improve=0.009250934, (0 missing)
## Surrogate splits:
##      children < 0.5 to the right, agree=0.643, adj=0.145, (0 split)
##      bmi < 20.805 to the left, agree=0.588, adj=0.014, (0 split)
##
## Node number 22: 145 observations
## mean=9.180171, MSE=0.09394768
##
## Node number 23: 202 observations
## mean=9.493345, MSE=0.08321284
```

```
# Plots tree
rpart.plot(rt1)
```



```
# Calculates rmse on training data
pred <- exp(predict(rt1, newdata=df_train))

sqrt(mean((df[as.numeric(names(pred)),$charges - pred)^2))
```

```
## [1] 4935.527
```

```
# Calculates rmse of testing data
pred <- exp(predict(rt1, newdata=df_test))
```

```
sqrt(mean((df_test$charges - pred)^2))
```

```
## [1] 4655.858
```

```
# Creates more complex tree with all predictors
```

```
rt2 <- rpart(log_charges ~ smoker + age + bmi + children + region + sex,  
             data=df_train, method="anova",  
             control = rpart.control(cp = 0.001))
```

```
# Prints tree summary
```

```
summary(rt2)
```

```
## Call:
```

```
## rpart(formula = log_charges ~ smoker + age + bmi + children +  
##       region + sex, data = df_train, method = "anova", control = rpart.control(cp = 0.001))  
## n= 936
```

```
##  
##          CP nsplit rel error    xerror      xstd  
## 1  0.433819896      0 1.0000000 1.0032355 0.03873227  
## 2  0.259251664      1 0.5661801 0.5676569 0.02488483  
## 3  0.042346919      2 0.3069284 0.3261008 0.02225331  
## 4  0.033533050      3 0.2645815 0.2762349 0.02196791  
## 5  0.029907101      4 0.2310485 0.2670372 0.02200265  
## 6  0.010757423      5 0.2011414 0.2255101 0.02178798  
## 7  0.008568249      6 0.1903839 0.2142417 0.02206472  
## 8  0.008212375      7 0.1818157 0.2081315 0.02216035  
## 9  0.007003669      8 0.1736033 0.2004927 0.02154070  
## 10 0.003521744      9 0.1665997 0.1878692 0.02113880  
## 11 0.002871287     11 0.1595562 0.1895375 0.02185439  
## 12 0.002650693     12 0.1566849 0.1874827 0.02181564  
## 13 0.002633144     13 0.1540342 0.1879872 0.02185405  
## 14 0.002346581     14 0.1514010 0.1875860 0.02200927  
## 15 0.001962506     15 0.1490545 0.1907078 0.02251759  
## 16 0.001957805     16 0.1470920 0.1927083 0.02337824  
## 17 0.001783203     18 0.1431763 0.1929762 0.02357001  
## 18 0.001652164     20 0.1396099 0.1907313 0.02355952  
## 19 0.001381187     21 0.1379578 0.1927346 0.02375809  
## 20 0.001377473     22 0.1365766 0.1922974 0.02363441  
## 21 0.001065842     23 0.1351991 0.1895143 0.02318785  
## 22 0.001043205     24 0.1341333 0.1873707 0.02309906  
## 23 0.001038963     25 0.1330901 0.1873707 0.02309906  
## 24 0.001019993     26 0.1320511 0.1873707 0.02309906  
## 25 0.001000000     27 0.1310311 0.1850103 0.02275716
```

```
##
```

```
## Variable importance
```

```
##   smoker    age    bmi children  region    sex  
##     48     39     5      5      3      1  
##
```

```
## Node number 1: 936 observations,    complexity param=0.4338199
```

```
##   mean=9.115445, MSE=0.8221967
```

```
##   left son=2 (747 obs) right son=3 (189 obs)
```

```
##   Primary splits:
```

```

##      smoker  splits as LR,          improve=0.433819900, (0 missing)
##      age    < 35.5    to the left, improve=0.226033000, (0 missing)
##      children < 1.5    to the left, improve=0.027968210, (0 missing)
##      bmi    < 35.88   to the left, improve=0.016811030, (0 missing)
##      region  splits as RLLL,        improve=0.001453492, (0 missing)
##
## Node number 2: 747 observations,    complexity param=0.2592517
## mean=8.815036, MSE=0.544556
## left son=4 (268 obs) right son=5 (479 obs)
## Primary splits:
##      age    < 32.5    to the left, improve=0.490467200, (0 missing)
##      children < 1.5    to the left, improve=0.055522360, (0 missing)
##      bmi    < 35.88   to the left, improve=0.016683860, (0 missing)
##      region  splits as RRLl,        improve=0.013072950, (0 missing)
##      sex     splits as RL,          improve=0.006867174, (0 missing)
## Surrogate splits:
##      bmi < 17.83    to the left, agree=0.645, adj=0.011, (0 split)
##
## Node number 3: 189 observations,    complexity param=0.0299071
## mean=10.30278, MSE=0.1530972
## left son=6 (89 obs) right son=7 (100 obs)
## Primary splits:
##      bmi    < 30.1    to the left, improve=0.795420500, (0 missing)
##      age    < 43.5    to the left, improve=0.110830400, (0 missing)
##      region  splits as LLRR,        improve=0.026738610, (0 missing)
##      children < 1.5    to the left, improve=0.007778093, (0 missing)
##      sex     splits as LR,          improve=0.004357908, (0 missing)
## Surrogate splits:
##      region  splits as LLRR,        agree=0.598, adj=0.146, (0 split)
##      sex     splits as LR,          agree=0.540, adj=0.022, (0 split)
##      age    < 21.5    to the left, agree=0.534, adj=0.011, (0 split)
##      children < 2.5    to the right, agree=0.534, adj=0.011, (0 split)
##
## Node number 4: 268 observations,    complexity param=0.03353305
## mean=8.124117, MSE=0.4624146
## left son=8 (143 obs) right son=9 (125 obs)
## Primary splits:
##      children < 0.5    to the left, improve=0.208237200, (0 missing)
##      age    < 22.5    to the left, improve=0.149463600, (0 missing)
##      region  splits as RRLl,        improve=0.049802280, (0 missing)
##      bmi    < 23.46   to the left, improve=0.018358330, (0 missing)
##      sex     splits as RL,          improve=0.008997053, (0 missing)
## Surrogate splits:
##      age    < 24.5    to the left, agree=0.664, adj=0.280, (0 split)
##      region  splits as LRLR,        agree=0.552, adj=0.040, (0 split)
##      sex     splits as LR,          agree=0.552, adj=0.040, (0 split)
##      bmi    < 28.215  to the left, agree=0.545, adj=0.024, (0 split)
##
## Node number 5: 479 observations,    complexity param=0.04234692
## mean=9.201604, MSE=0.1739923
## left son=10 (132 obs) right son=11 (347 obs)
## Primary splits:
##      age    < 42.5    to the left, improve=0.391028000, (0 missing)
##      bmi    < 34.785  to the left, improve=0.020083240, (0 missing)

```

```

##      sex      splits as  RL,          improve=0.012344180, (0 missing)
##      region   splits as  RRLl,        improve=0.007727859, (0 missing)
##      children < 0.5      to the right, improve=0.007035719, (0 missing)
##      Surrogate splits:
##      bmi < 18.0975 to the left, agree=0.729, adj=0.015, (0 split)
##
## Node number 6: 89 observations,      complexity param=0.002871287
##      mean=9.932874, MSE=0.04433209
##      left son=12 (49 obs) right son=13 (40 obs)
##      Primary splits:
##      age      < 41        to the left, improve=0.560041200, (0 missing)
##      bmi      < 22.74     to the left, improve=0.169516000, (0 missing)
##      sex      splits as  LR,          improve=0.015132370, (0 missing)
##      children < 0.5      to the left, improve=0.013474440, (0 missing)
##      region   splits as  LRLl,        improve=0.003564698, (0 missing)
##      Surrogate splits:
##      bmi < 27.02   to the right, agree=0.573, adj=0.05, (0 split)
##
## Node number 7: 100 observations,      complexity param=0.001038963
##      mean=10.63199, MSE=0.01974027
##      left son=14 (56 obs) right son=15 (44 obs)
##      Primary splits:
##      age      < 41.5      to the left, improve=0.40504080, (0 missing)
##      bmi      < 34.4575   to the left, improve=0.11190850, (0 missing)
##      sex      splits as  RL,          improve=0.02277753, (0 missing)
##      region   splits as  LRLl,        improve=0.02011163, (0 missing)
##      children < 0.5      to the left, improve=0.01603115, (0 missing)
##      Surrogate splits:
##      bmi      < 30.7325   to the right, agree=0.59, adj=0.068, (0 split)
##      region   splits as  LRLl,        agree=0.59, adj=0.068, (0 split)
##      children < 2.5      to the left, agree=0.57, adj=0.023, (0 split)
##
## Node number 8: 143 observations,      complexity param=0.008212375
##      mean=7.833995, MSE=0.3118515
##      left son=16 (80 obs) right son=17 (63 obs)
##      Primary splits:
##      age      < 22.5      to the left, improve=0.14172170, (0 missing)
##      sex      splits as  RL,          improve=0.05924124, (0 missing)
##      region   splits as  RRLl,        improve=0.03715225, (0 missing)
##      bmi      < 31.0975   to the right, improve=0.02103288, (0 missing)
##      Surrogate splits:
##      bmi      < 28.025    to the right, agree=0.615, adj=0.127, (0 split)
##      region   splits as  LRLl,        agree=0.587, adj=0.063, (0 split)
##
## Node number 9: 125 observations,      complexity param=0.008568249
##      mean=8.456018, MSE=0.4282089
##      left son=18 (68 obs) right son=19 (57 obs)
##      Primary splits:
##      region   splits as  RRLl,        improve=0.123190700, (0 missing)
##      age      < 18.5      to the left, improve=0.073362980, (0 missing)
##      children < 1.5      to the left, improve=0.071804430, (0 missing)
##      bmi      < 23.46     to the left, improve=0.028392590, (0 missing)
##      sex      splits as  RL,          improve=0.006242444, (0 missing)
##      Surrogate splits:

```

```

##      bmi      < 29.3625 to the right, agree=0.648, adj=0.228, (0 split)
##      children < 1.5      to the left,  agree=0.560, adj=0.035, (0 split)
##
## Node number 10: 132 observations,      complexity param=0.002346581
## mean=8.778695, MSE=0.09123539
## left son=20 (76 obs) right son=21 (56 obs)
## Primary splits:
##      children < 1.5      to the left,  improve=0.1499512000, (0 missing)
##      region  splits as  RLLL,          improve=0.0796782500, (0 missing)
##      age     < 36.5      to the left,  improve=0.0376981000, (0 missing)
##      bmi     < 35.0075 to the left,  improve=0.0327415000, (0 missing)
##      sex     splits as  LR,           improve=0.0003814494, (0 missing)
## Surrogate splits:
##      bmi     < 23.5125 to the right, agree=0.621, adj=0.107, (0 split)
##      age     < 33.5      to the right, agree=0.598, adj=0.054, (0 split)
##      region splits as  RLLL,          agree=0.591, adj=0.036, (0 split)
##
## Node number 11: 347 observations,      complexity param=0.01075742
## mean=9.36248, MSE=0.1115564
## left son=22 (145 obs) right son=23 (202 obs)
## Primary splits:
##      age     < 51.5      to the left,  improve=0.213863100, (0 missing)
##      sex     splits as  RL,           improve=0.023005920, (0 missing)
##      bmi     < 23.6275 to the left,  improve=0.017386360, (0 missing)
##      region splits as  RRLl,          improve=0.011560880, (0 missing)
##      children < 2.5      to the left,  improve=0.009250934, (0 missing)
## Surrogate splits:
##      children < 0.5      to the right, agree=0.643, adj=0.145, (0 split)
##      bmi     < 20.805   to the left,  agree=0.588, adj=0.014, (0 split)
##
## Node number 12: 49 observations
## mean=9.79051, MSE=0.01373823
##
## Node number 13: 40 observations
## mean=10.10727, MSE=0.02656772
##
## Node number 14: 56 observations
## mean=10.55273, MSE=0.01225899
##
## Node number 15: 44 observations
## mean=10.73287, MSE=0.01109006
##
## Node number 16: 80 observations,      complexity param=0.002650693
## mean=7.647435, MSE=0.3310694
## left son=32 (56 obs) right son=33 (24 obs)
## Primary splits:
##      region splits as  RLLL,          improve=0.07701972, (0 missing)
##      sex     splits as  RL,           improve=0.06548628, (0 missing)
##      bmi     < 28.7425 to the left,  improve=0.03686928, (0 missing)
##      age     < 18.5      to the right, improve=0.01773498, (0 missing)
## Surrogate splits:
##      age < 18.5      to the right, agree=0.713, adj=0.042, (0 split)
##
## Node number 17: 63 observations,      complexity param=0.001783203

```



```

## mean=8.070896, MSE=0.1871295
## left son=34 (39 obs) right son=35 (24 obs)
## Primary splits:
##   age < 27.5 to the left, improve=0.10982070, (0 missing)
##   sex splits as RL, improve=0.10039900, (0 missing)
##   region splits as RRLR, improve=0.05240119, (0 missing)
##   bmi < 33.905 to the right, improve=0.03262175, (0 missing)
## Surrogate splits:
##   bmi < 20.66 to the right, agree=0.635, adj=0.042, (0 split)
##
## Node number 18: 68 observations, complexity param=0.007003669
## mean=8.245737, MSE=0.2792275
## left son=36 (12 obs) right son=37 (56 obs)
## Primary splits:
##   age < 19.5 to the left, improve=0.28386380, (0 missing)
##   bmi < 35.255 to the left, improve=0.05933584, (0 missing)
##   children < 1.5 to the left, improve=0.03096690, (0 missing)
##   sex splits as RL, improve=0.02802208, (0 missing)
##   region splits as --RL, improve=0.01604467, (0 missing)
##
## Node number 19: 57 observations, complexity param=0.003521744
## mean=8.706879, MSE=0.4902582
## left son=38 (21 obs) right son=39 (36 obs)
## Primary splits:
##   children < 1.5 to the left, improve=0.0919267900, (0 missing)
##   bmi < 26.3625 to the left, improve=0.0569415500, (0 missing)
##   region splits as RL--, improve=0.0257846300, (0 missing)
##   age < 29.5 to the right, improve=0.0097739930, (0 missing)
##   sex splits as LR, improve=0.0001898478, (0 missing)
## Surrogate splits:
##   age < 19.5 to the left, agree=0.684, adj=0.143, (0 split)
##   bmi < 37.24 to the right, agree=0.667, adj=0.095, (0 split)
##
## Node number 20: 76 observations, complexity param=0.001065842
## mean=8.678292, MSE=0.05412017
## left son=40 (32 obs) right son=41 (44 obs)
## Primary splits:
##   age < 36.5 to the left, improve=0.1994214000, (0 missing)
##   bmi < 34.96 to the left, improve=0.0885448200, (0 missing)
##   region splits as RRRRL, improve=0.0254618000, (0 missing)
##   children < 0.5 to the left, improve=0.0040481320, (0 missing)
##   sex splits as LR, improve=0.0004014217, (0 missing)
## Surrogate splits:
##   bmi < 23.75 to the left, agree=0.645, adj=0.156, (0 split)
##
## Node number 21: 56 observations, complexity param=0.001043205
## mean=8.914955, MSE=0.1093583
## left son=42 (37 obs) right son=43 (19 obs)
## Primary splits:
##   region splits as RLLL, improve=0.131093500, (0 missing)
##   age < 33.5 to the left, improve=0.082377080, (0 missing)
##   children < 3.5 to the left, improve=0.072171460, (0 missing)
##   bmi < 37.715 to the left, improve=0.030499340, (0 missing)
##   sex splits as RL, improve=0.001527673, (0 missing)

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## Surrogate splits:
##     bmi < 23.8925 to the right, agree=0.804, adj=0.421, (0 split)
##
## Node number 22: 145 observations
##     mean=9.180171, MSE=0.09394768
##
## Node number 23: 202 observations,     complexity param=0.001962506
##     mean=9.493345, MSE=0.08321284
##     left son=46 (132 obs) right son=47 (70 obs)
##     Primary splits:
##         age < 58.5 to the left, improve=0.08985058, (0 missing)
##         children < 1.5 to the left, improve=0.03622791, (0 missing)
##         sex splits as RL, improve=0.02093293, (0 missing)
##         bmi < 23.865 to the left, improve=0.01709518, (0 missing)
##         region splits as RRL, improve=0.01701240, (0 missing)
##
## Node number 32: 56 observations,     complexity param=0.001957805
##     mean=7.542898, MSE=0.2699072
##     left son=64 (24 obs) right son=65 (32 obs)
##     Primary splits:
##         sex splits as RL, improve=0.07823523, (0 missing)
##         age < 20.5 to the left, improve=0.03805859, (0 missing)
##         region splits as -RRL, improve=0.03262846, (0 missing)
##         bmi < 23.955 to the left, improve=0.02335344, (0 missing)
##     Surrogate splits:
##         bmi < 39.495 to the right, agree=0.607, adj=0.083, (0 split)
##         age < 20.5 to the right, agree=0.589, adj=0.042, (0 split)
##
## Node number 33: 24 observations,     complexity param=0.002633144
##     mean=7.891356, MSE=0.388785
##     left son=66 (12 obs) right son=67 (12 obs)
##     Primary splits:
##         bmi < 28.975 to the left, improve=0.21717280, (0 missing)
##         sex splits as RL, improve=0.04051000, (0 missing)
##         age < 19 to the right, improve=0.01553318, (0 missing)
##     Surrogate splits:
##         sex splits as RL, agree=0.792, adj=0.583, (0 split)
##         age < 19 to the right, agree=0.583, adj=0.167, (0 split)
##
## Node number 34: 39 observations,     complexity param=0.001783203
##     mean=7.958439, MSE=0.1855495
##     left son=68 (20 obs) right son=69 (19 obs)
##     Primary splits:
##         sex splits as RL, improve=0.200365000, (0 missing)
##         bmi < 29.0425 to the right, improve=0.108208000, (0 missing)
##         region splits as RRLR, improve=0.093994690, (0 missing)
##         age < 24.5 to the right, improve=0.005352531, (0 missing)
##     Surrogate splits:
##         age < 24.5 to the right, agree=0.615, adj=0.211, (0 split)
##         region splits as LRLR, agree=0.615, adj=0.211, (0 split)
##         bmi < 29.0425 to the right, agree=0.564, adj=0.105, (0 split)
##
## Node number 35: 24 observations
##     mean=8.253638, MSE=0.1357515

```

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##
## Node number 36: 12 observations
##   mean=7.637549, MSE=0.08267421
##
## Node number 37: 56 observations,   complexity param=0.001377473
##   mean=8.376063, MSE=0.2250987
##   left son=74 (38 obs) right son=75 (18 obs)
##   Primary splits:
##     age      < 29.5    to the left, improve=0.084095710, (0 missing)
##     region    splits as  --RL,      improve=0.062425900, (0 missing)
##     bmi       < 35.255  to the left, improve=0.029254640, (0 missing)
##     children  < 1.5     to the right, improve=0.004768032, (0 missing)
##     sex       splits as  RL,        improve=0.002111917, (0 missing)
##   Surrogate splits:
##     bmi < 42.76   to the left, agree=0.696, adj=0.056, (0 split)
##
## Node number 38: 21 observations,   complexity param=0.003521744
##   mean=8.428924, MSE=0.3858636
##   left son=76 (14 obs) right son=77 (7 obs)
##   Primary splits:
##     region splits as  RL--,      improve=0.35191710, (0 missing)
##     age      < 23.5    to the left, improve=0.21361150, (0 missing)
##     bmi      < 27.5975 to the left, improve=0.10953330, (0 missing)
##     sex      splits as  LR,      improve=0.06380832, (0 missing)
##   Surrogate splits:
##     age < 26.5     to the left, agree=0.762, adj=0.286, (0 split)
##     bmi < 28.0725  to the left, agree=0.762, adj=0.286, (0 split)
##
## Node number 39: 36 observations,   complexity param=0.001381187
##   mean=8.86902, MSE=0.4797976
##   left son=78 (21 obs) right son=79 (15 obs)
##   Primary splits:
##     age      < 25.5    to the right, improve=0.061538020, (0 missing)
##     bmi      < 32.6325 to the left, improve=0.048197190, (0 missing)
##     sex      splits as  RL,      improve=0.018037060, (0 missing)
##     region    splits as  LR--,    improve=0.005250455, (0 missing)
##     children  < 2.5     to the right, improve=0.003912136, (0 missing)
##   Surrogate splits:
##     bmi      < 27.455  to the right, agree=0.667, adj=0.200, (0 split)
##     children  < 3.5     to the left, agree=0.611, adj=0.067, (0 split)
##
## Node number 40: 32 observations
##   mean=8.556473, MSE=0.05203085
##
## Node number 41: 44 observations
##   mean=8.766888, MSE=0.03699771
##
## Node number 42: 37 observations
##   mean=8.829154, MSE=0.04331039
##
## Node number 43: 19 observations
##   mean=9.082041, MSE=0.195724
##
## Node number 46: 132 observations

```

```

## mean=9.430377, MSE=0.07779665
##
## Node number 47: 70 observations, complexity param=0.001652164
## mean=9.612084, MSE=0.07185054
## left son=94 (49 obs) right son=95 (21 obs)
## Primary splits:
## children < 0.5 to the left, improve=0.252799800, (0 missing)
## bmi < 33.265 to the left, improve=0.036965570, (0 missing)
## sex splits as RL, improve=0.017565810, (0 missing)
## age < 60.5 to the left, improve=0.015080250, (0 missing)
## region splits as LRRl, improve=0.004732582, (0 missing)
## Surrogate splits:
## age < 59.5 to the right, agree=0.729, adj=0.095, (0 split)
##
## Node number 64: 24 observations, complexity param=0.001957805
## mean=7.375103, MSE=0.2780882
## left son=128 (15 obs) right son=129 (9 obs)
## Primary splits:
## age < 20.5 to the left, improve=0.27432090, (0 missing)
## bmi < 32.72 to the right, improve=0.13427260, (0 missing)
## region splits as -RLL, improve=0.03039037, (0 missing)
## Surrogate splits:
## region splits as -RLL, agree=0.667, adj=0.111, (0 split)
##
## Node number 65: 32 observations, complexity param=0.001019993
## mean=7.668743, MSE=0.2268181
## left son=130 (20 obs) right son=131 (12 obs)
## Primary splits:
## bmi < 33.59 to the left, improve=0.10814870, (0 missing)
## age < 18.5 to the right, improve=0.05781933, (0 missing)
## region splits as -RRL, improve=0.05080885, (0 missing)
## Surrogate splits:
## age < 18.5 to the right, agree=0.719, adj=0.250, (0 split)
## region splits as -LRL, agree=0.688, adj=0.167, (0 split)
##
## Node number 66: 12 observations
## mean=7.600781, MSE=0.02399664
##
## Node number 67: 12 observations
## mean=8.18193, MSE=0.5847063
##
## Node number 68: 20 observations
## mean=7.770506, MSE=0.02105956
##
## Node number 69: 19 observations
## mean=8.156263, MSE=0.2823849
##
## Node number 74: 38 observations
## mean=8.28137, MSE=0.2079973
##
## Node number 75: 18 observations
## mean=8.57597, MSE=0.2023087
##
## Node number 76: 14 observations

```

```

##   mean=8.168355, MSE=0.1716139
##
## Node number 77: 7 observations
##   mean=8.950061, MSE=0.4069869
##
## Node number 78: 21 observations
##   mean=8.723797, MSE=0.226091
##
## Node number 79: 15 observations
##   mean=9.072333, MSE=0.764125
##
## Node number 94: 49 observations
##   mean=9.523854, MSE=0.02250088
##
## Node number 95: 21 observations
##   mean=9.817953, MSE=0.1264538
##
## Node number 128: 15 observations
##   mean=7.161161, MSE=0.02047268
##
## Node number 129: 9 observations
##   mean=7.731673, MSE=0.5040196
##
## Node number 130: 20 observations
##   mean=7.547426, MSE=0.0131118
##
## Node number 131: 12 observations
##   mean=7.87094, MSE=0.5175817

```

```

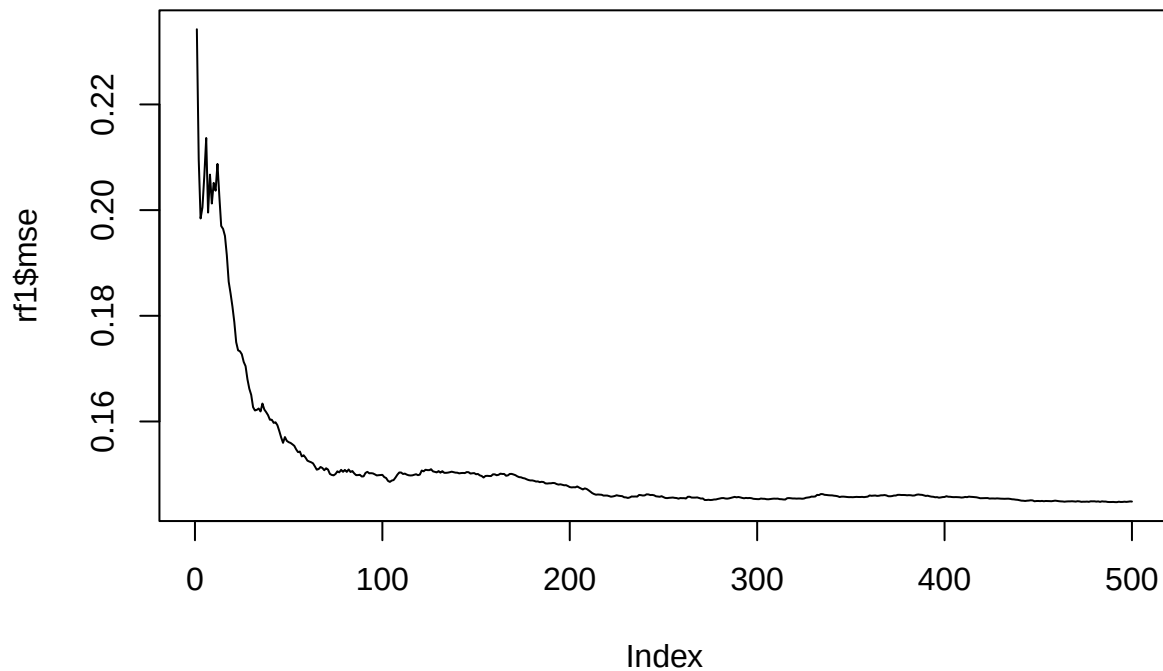
# Plots tree
rpart.plot(rt2)

```



```
## predicted      936    -none- numeric
## mse           500    -none- numeric
## rsq           500    -none- numeric
## oob.times      936    -none- numeric
## importance      12    -none- numeric
## importanceSD     6    -none- numeric
## localImportance  0    -none- NULL
## proximity       0    -none- NULL
## ntree           1    -none- numeric
## mtry            1    -none- numeric
## forest         11    -none- list
## coefs           0    -none- NULL
## y              936    -none- numeric
## test            0    -none- NULL
## inbag           0    -none- NULL
## terms           3    terms  call
```

```
# Plots error over number of trees
plot(rf1$mse, type = "l")
```



```
# Calculates training rmse
pred <- exp(predict(rf1, newdata=df_train))

sqrt(mean((df[as.numeric(names(pred)),]$charges - pred)^2))
```

```
## [1] 3740.856
```

```

# Calculates testing rmse
pred <- exp(predict(rf1, newdata=df_test))

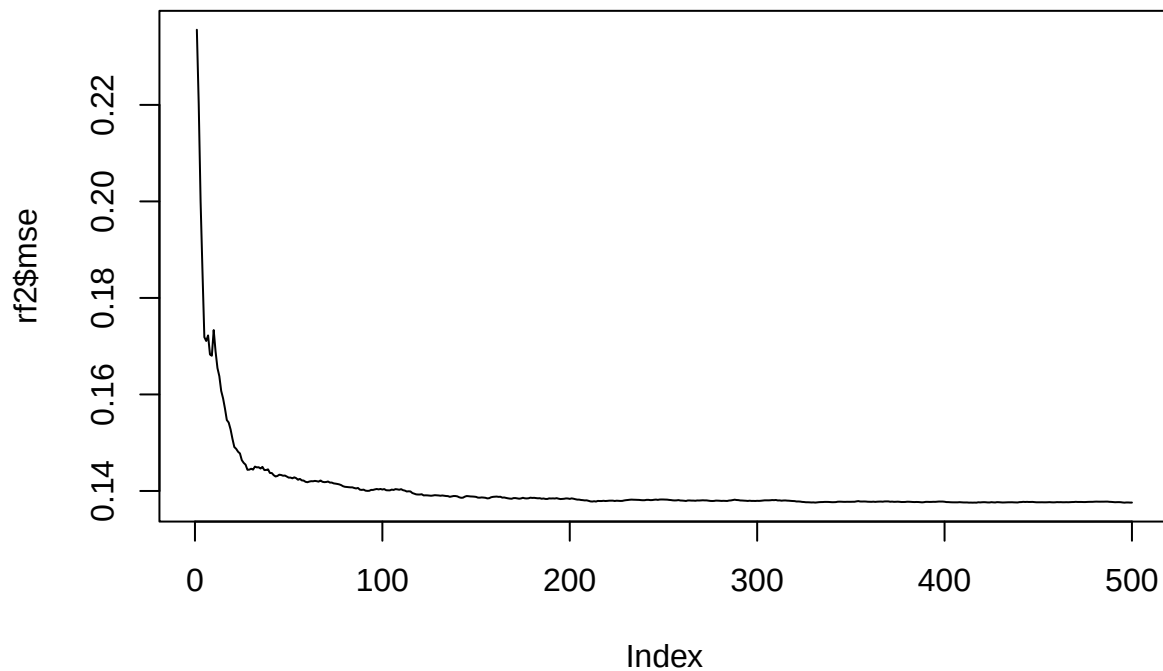
sqrt(mean((df_test$charges - pred)^2))

## [1] 4615.851

# Creates random forest with different parameters
rf2 <- randomForest(log_charges ~ smoker + bmi + age + children + region + sex,
                    data=df_train, ntree=500, mtry=4, nodesize=27)

# Plots mse over number of trees
plot(rf2$mse, type = "l")

```



```

# Calculates rmse for training set
pred <- exp(predict(rf2, newdata=df_train))

sqrt(mean((df[as.numeric(names(pred)),$charges - pred)^2))

## [1] 4017.597

# Calculates rmse for testing set
pred <- exp(predict(rf2, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))

```


[1] 4221.199